

# ANANDHAN ANBALAGAN

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## SUMMARY

Lead Software Engineer with 8+ years of expertise in software development, complex algorithm design, and scalable system architecture. Adept at driving end-to-end data integration and migration strategies across cloud environments using Azure Data Factory, Informatica, and robust DevOps/CI/CD workflows. A methodical problem-solver recognized for optimizing system performance, streamlining legacy integrations, and establishing automated task scheduling to eliminate manual interventions.

## SKILLS

**Cloud & ETL:** Azure Data Factory (ADF), Informatica IICS, IICS CDI, IICS CAI, DataStage, Data Integration, Cloud Data Migration, ETL Architecture

**Tools & Databases:** SQL, SAP ECC, Salesforce REST APIs, Automic UC4, Git, GitHub, Visual Studio

**DevOps & Methodologies:** Agile/Scrum, CI/CD, Azure DevOps, Atlassian Jira, Confluence, Requirements Gathering, Architecture Design, Technical Design Documents

## EXPERIENCE

### LEAD SOFTWARE ENGINEER | ECOLAB DIGITAL CENTER

May 2021 - Present | Bengaluru

#### Project APS (January 2024 - Present)

- Requirement Gathering: Partnered with stakeholders to capture business needs and convert them into actionable user stories for upcoming feature development, leveraging Agile methodologies and requirements-gathering tools.
- Oversaw end-to-end integration projects to facilitate seamless data migration into SAP ECC by coordinating efforts between stakeholders and software teams, incorporating ETL technologies.
- Data Integrity: Collaborated with product owners and QA teams to design and implement advanced validation logic in SQL ETL workflows, reducing data ingestion errors by 95%.
- Performance Optimization: Designed and executed Informatica Cloud (IICS) integration patterns using parallel processing techniques, improving SAP ECC IDOC processing throughput by 30%.
- DevOps Architecture: Established Git branching and release strategies to facilitate smooth cross-environment job migrations via automated CI/CD deployment pipelines.
- Process Automation: Engineered end-to-end task automation workflows leveraging UC4 scheduling software and scripting languages, eliminating manual operational interventions.
- Mentored junior developers by reviewing code and providing feedback on architectural decisions, utilizing software design patterns and modern development frameworks.
- Examined and validated technical documentation to ensure alignment with industry compliance standards through targeted review processes.

### SENIOR SOFTWARE ENGINEER | ECOLAB DIGITAL CENTER

May 2021 - Present

#### Project Connect Migration (January 2022 - January 2024)

- Strategic Planning: Translated complex business requirements and technical system constraints into comprehensive architecture design and migration strategy proof-of-concepts (POCs).
- Translated business requirements and technical constraints into detailed design documents, applying knowledge of design principles and leveraging modeling tools.
- Engineered robust data movement and transformation processes using Azure Data Factory by configuring linked services, datasets, pipelines, and dataflows to support enterprise-scale data operations.
- Engineered error and log file management systems using scripting languages and cloud storage solutions to optimize retrieval processes.
- Cloud Infrastructure: Engineered enterprise-scale data movement and transformation processes by configuring modular Azure Data Factory (ADF) linked services, datasets, and pipelines.
- API Integration: Designed and deployed a reusable ADF pipeline architecture to execute Salesforce data deletion tasks via REST APIs, establishing scalable data operation patterns.
- Directed the resolution of 50+ complex integration challenges over a 2-year span using advanced ADF troubleshooting techniques and SQL-based diagnostics, improving pipeline reliability by 35%.
- Implemented systematic debugging protocols across Azure Data Factory pipelines and integrated SQL validation scripts from January 2022 to January 2024, reducing data processing errors by 40% and accelerating error resolution turnaround time within a large-scale migration project involving over 100 data flows.

### SOFTWARE ENGINEER | ECOLAB DIGITAL CENTER

May 2021 - Present

#### Project Connect (May 2021 - January 2022)

- Legacy Modernization: Enhanced legacy data integrations using Informatica Cloud Services (IICS), significantly increasing system reliability and maintainability between critical enterprise platforms.
- Pipeline Design: Guided the end-to-end design of ETL workflows utilizing Mapping Tasks, MCTs, and Task Flows to streamline cloud-based data operations.
- Executed performance analysis and tuning of critical system components, contributing to increased throughput and reliability across distributed services.
- Documentation & Onboarding: Standardized mapping sheets and integration guides with structured versioning on cloud documentation platforms, drastically reducing developer project ramp-up times.

**ASSOCIATE SOFTWARE ENGINEER | UST****January 2018 - May 2021 |****Trivandrum**

- Created step-by-step process guides by analyzing job workflows and capturing essential procedures with documentation software.
- Streamlined data integration workflows by enhancing ETL pipelines with advanced automation scripts, resulting in significantly improved job execution reliability and resource utilization.
- Workflow Automation: Streamlined backend data integration workflows by embedding advanced automation scripts into legacy ETL pipelines, boosting job execution reliability.

**EDUCATION**

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Bachelor of Engineering, Electronics and Communication Engineering | SNS College of Technology, Coimbatore | 2016 | CGPA: 7.42